

Question Examples

Let f, g be the functions defined as $f(n) = 10n \log n$ and $g(n) = \frac{1}{n} \log n$.

Scegli una o più alternative:

- ☐ $f \in \Theta(g)$
- ☐ $f \in O(g)$
- ☒ $f \in \Omega(g)$

$$\lim_{n \rightarrow +\infty} \frac{10n \log n}{\frac{1}{n} \log n} = \lim_{n \rightarrow +\infty} 10n^2 = +\infty \quad f \in \Omega(g)$$

Turing Machines:

Scegli una o più alternative:

- ☒ Are such that certain problems that can be computed in time n^2 cannot be computed in time n
- ☒ Can contain an arbitrary number of work tapes 
- ☐ All work in exponential time
- ☐ All work in polynomial time

The notion of PAC-learnable concept class:

Scegli una o più alternative:

- ☐ Cannot be reached when the underlying concept class is the one conjunctions of literals.
- ☒ Needs to hold for every distribution $\mathbf{D}$ on the instance class.
- ☒ Requires the output concept to have probability of error ϵ , in all cases 
- ☒ Does not make any reference to the time complexity of the learning algorithm 

Let f, g be the functions defined as $f(n) = 2^{10} n^2 \log^2 n$ and $g(n) = 10^{-3} n^3 \log n$.

Scegli una o più alternative:

- ☐ $f \in \Theta(g)$
- ☐ $f \in \Omega(g)$
- ☒ $f \in O(g)$

$$\lim_{n \rightarrow +\infty} \frac{2^{10} n^2 \log^2 n}{10^{-3} n^3 \log n} = \frac{2^{10} \cdot 2^{10} \cdot \log n}{n} = 0 \quad f \in O(g)$$

In Turing Machines:

Scegli una o più alternative:

- ☒ What can be computed in exponential time is different from what can be computed in polynomial time
- ☐ The class **EXP** can be equal to **P**
- ☒ The presence of many tapes can make the class **DTIME**(n) different
- ☐ The presence of many tapes can make the class **P** different

Which one(s) among the following reductions hold:

Scegli una o più alternative:

- ☒ **CLIQUE** $\leq_p$ **SAT**
- ☒ **SAT** $\leq_p$ **SAT**
- ☐ **INDSET** $\leq_p$ **2SAT**
- ☒ **2SAT** $\leq_p$ **INDSET**

Suppose a language $\mathcal{L}$ is both in **NP** and in **EXP**. Then

Scegli una o più alternative:

- ☒ The classes **EXP** and **NP** are maybe different.
- ☒ $\mathcal{L}$ can even be **NP**-complete
- ☐ The classes **EXP** and **NP** are necessarily equal.
- ☐ $\mathcal{L}$ cannot be in **P**

Let f, g be the functions defined as $f(n) = n^2 2^n$ and $g(n) = n 2^n$.

Scegli una o più alternative:

- ☐ $f \in \Theta(g)$
- ☒ $f \in \Omega(g)$
- ☐ $f \in O(g)$

$$\lim_{n \rightarrow \infty} \frac{n^2 2^n}{n 2^n} = +\infty \quad f(n) = \Omega(g(n))$$

The problem **3SAT** is:

Scegli una o più alternative:

- ☒ such that **INDSET** can be reduced to it, i.e., $\text{INDSET} \leq_p \text{3SAT}$
- ☒ **NP**-hard
- ☒ In the class **EXP**.
- ☐ Computable in polynomial time

Suppose a language $\mathcal{L}$ is both in **P** and **NP**-complete. Then:

Scegli una o più alternative:

- ☒ The classes **EXP** and **P** are different.
- ☐ The classes **P** and **NP** are different.
- ☒ The classes **P** and **NP** are equal.
- ☐ The classes **EXP** and **NP** are equal

The halting problem is:

Scegli una o più alternative:

- ☒ Undecidable, but **NP**-hard.
- ☐ Nothing about its computational complexity is known
- ☐ **NP**-complete
- ☐ Decidable in polynomial time

Let f, g be the functions defined as $f(n) = n2^{-15}n^2 \log n$ and $g(n) = \frac{2^{30}}{\log^{-1} n}n^3$.

Scegli una o più alternative:

- ☒ $f \in \Theta(g)$
- ☐ $f \notin O(g)$
- ☒ $f \in \Omega(g)$

✓

✓

Le risposte corrette sono: $f \in \Omega(g), f \in \Theta(g)$

$$\lim_{n \rightarrow +\infty} \frac{n2^{-15}n^2 \log n}{\frac{2^{30}}{\log^{-1} n}n^3} = \frac{\cancel{n^3} \log n \cdot \log^{-1} n}{2^{45} \cancel{n^3}} = \frac{1}{2^{45}} \quad f \in \Theta(g)$$

Turing Machines:

Scegli una o più alternative:

- ☐ Have finitely many configurations.
- ☒ Can have as many tapes as needed.
- ☐ Always halt their execution on any input.
- ☒ Have finitely many states

✓

✓

Le risposte corrette sono: Can have as many tapes as needed., Have finitely many states

Which of the following assertions is true about the problem of checking whether two sequences of natural numbers contain exactly the same elements (both of them considered as part of the input)?

Scegli una o più alternative:

- ☒ It is in **EXP**
- ☒ It can be polynomially reduced to the 3SAT problem.
- ☒ It is in **P**
- ☐ It is well-known to be **NP**-hard

✓

✓

Le risposte corrette sono: It is in **P**, It is in **EXP**, It can be polynomially reduced to the 3SAT problem.

Which ones of the following inequalities between complexity classes are currently known to be true? Here, **NPC** stands for the class of all **NP**-complete problems.

Scegli una o più alternative:

- ☐ **EXP** $\neq$ **NPC**.
- ☒ **P** $\neq$ **NPC**.
- ☒ **EXP** $\neq$ **P**.
- ☐ **NP** $\neq$ **NPC**.

✗

✓

La risposta corretta è: **EXP** $\neq$ **P**.

The concept class consisting of conjunctions of literals:

Scegli una o più alternative:

- ☐ Is not efficiently PAC-learnable, provided **RP** $\neq$ **NP**.
- ☐ Has to do with rectangles.
- ☒ Is efficiently PAC-learnable.
- ☒ Is easier to learn than the concept class of CNFs.

✓

Le risposte corrette sono: Is efficiently PAC-learnable., Is easier to learn than the concept class of CNFs.

Let f, g be the functions defined as $f(n) = 10^{-5}n^{10} \log n$ and $g(n) = \frac{n^{10}}{\log n}$.

Scegli una o più alternative:

- ☐ $f \in O(g)$
- ☐ $f \in \Theta(g)$
- ☒ $f \in \Omega(g)$

✓

Turing Machines:

Scegli una o più alternative:

- ☒ Are such that certain problems that can be computed in time 2^n cannot be computed in time n^2
- ☒ Can work on alphabets of arbitrary but fixed size
- ☐ All work in time bounded by a computable function
- ☐ All work in time bounded by a polynomial function

✓

✓

The problem of deciding if a 3CNF is satisfiable is

Scegli una o più alternative:

- ☐ In the class **P**
- ☒ In the class **EXP**
- ☒ Undecidable, but **NP**-hard.
- ☒ **NP**-complete

✓

✗

✓

Suppose a language $\mathcal{L}$ is both undecidable and not in **NP**. Then:

Scegli una o più alternative:

- ☐ The classes **P** and **NP** are necessarily equal.
- ☐ The classes **P** and **NP** are necessarily different.
- ☒ The language $\mathcal{L}$ is strictly harder to decide than **3SAT**
- ☒ There cannot be any deterministic Turing machine deciding $\mathcal{L}$



The notion of PAC-learnable concept class:

Scegli una o più alternative:

- ☐ Cannot be reached when the underlying concept class is a conjunctions of literals.
- ☒ Is such that the accuracy and confidence errors are taken into account independently.
- ☒ Implicitly refers to the underlying representation class.
- ☐ Is such that the output concept needs to have probability of error ϵ , in all possible cases



Let f, g be the functions defined as $f(n) = 10^3 n \log n$ and $g(n) = \frac{n^2}{10^5 \log n}$.

Scegli una o più alternative:

- ☐ $f \in \Omega(g)$
- ☐ $f \in \Theta(g)$
- ☒ $f \in O(g)$

Nondeterministic Turing Machines:

Scegli una o più alternative:

- ☒ Can be simulated by deterministic TMs.
- ☒ If working in polynomial time, can be used to characterize **NP**
- ☐ Always work in polynomial time
- ☐ Are essential to define the complexity class **NP**

The universal Turing machine:

Scegli una o più alternative:

- ☒ Can simulate every Turing machine, with a polynomial overhead.
- ☐ Can simulate every Turing machine, but not itself
- ☐ Works in polynomial time.
- ☒ Is an essential ingredient of in the proof of existence of uncomputable problems.

Suppose a language $\mathcal{L}$ is in **EXP** but not in **P**. Then:

Scegli una o più alternative:

- ☐ $\mathcal{L}$ is necessarily **NP**-complete.
- ☒ The classes **NP** and **P** are different.
- ☒ There could be a nondeterministic polytime TM computing $\mathcal{L}$
- ☐ $\mathcal{L}$ can be computed in polynomial time.

The notion of PAC-learnable concept class:

Select one or more:

- ☒ Needs to hold for every distribution **D** on the instance class.
- ☐ Cannot be reached when the underlying concept class is the one conjunctions of literals.
- ☐ Requires the output concept to have probability of error ε , in all cases
- ☒ Does not make any reference to the time complexity of the learning algorithm

In Turing Machines:

Select one or more:

- ☒ The presence of many tapes can make the class **DTIME**(n) different
- ☒ What can be computed in exponential time is different from what can be computed in polynomial time
- ☐ The presence of many tapes can make the class **P** different
- ☐ The class **EXP** can be equal to **P**

Let f, g be the functions defined as $f(n) = 10^{-5}n^{10} \log n$ and $g(n) = \frac{n^{10}}{\log n}$.

Scegli una o più alternative:

- ☐ $f \in O(g)$
- ☐ $f \in \Theta(g)$
- ☒ $f \in \Omega(g)$



$$\lim_{n \rightarrow +\infty} \frac{10^{-5} n^{10} \log n}{\frac{n^{10}}{\log n}} = \frac{10^{-5} \cancel{n^{10}} \log^2 n}{\cancel{n^{10}}} = +\infty \quad f \in \Omega(g)$$

The concept classes of axis-aligned squares in $\mathbb{R}^2$ is:

.

Scegli una o più alternative:

- ☐ Not even PAC learnable
- ☒ PAC learnable
- ☒ Efficiently PAC learnable.
- ☐ PAC learnable, but not efficiently so.



The problem of checking whether a Turing machine rejects the string 00101 is:

Scegli una o più alternative:

- ☐ In the class **NP**
- ☒ Not in the class **P** 
- ☐ Decidable in linear time
- ☒ Undecidable. 

Suppose a friend of yours come to you saying he found a language $\mathcal{L}$ such that $\text{SAT} \subseteq \mathcal{L}$ and which is decidable in polynomial time.

You can conclude that:

Scegli una o più alternative:

- ☐ Your friend cannot be right.
- ☐ Your friend has proved that $\mathbf{P} \neq \mathbf{NP}$.
- ☒ Your friend is possibly right. 
- ☒ Your friend has proved that $\mathbf{P} = \mathbf{NP}$. 

Let f, g be the

functions defined as

$$f(n) = n10^4 n^n \log^3 n$$

and

$$g(n) = 10^5 n^{n+1} \log^3 n.$$

Scegli una o più alternative:

☒ $f \in \Omega(g)$

☒ $f \in \Theta(g)$

☒ $f \in O(g)$



$$\lim_{n \rightarrow +\infty} \frac{n10^4 n^n \log^3 n}{10^5 n^{n+1} \log^3 n} =$$

$$\lim_{n \rightarrow +\infty} \frac{10^4}{10^5}$$

$$f \in \Theta(g)$$

Which of the following sets can be easily encoded as binary strings?

Scegli una o più alternative:

- ☒ Finite graphs ✓
- ☐ Infinite sequences of real numbers
- ☒ Natural numbers ✓
- ☒ Finite sequences of real numbers ✗

The problem 2SAT is:

Scegli una o più alternative:

- ☐ such that INDSET can be reduced to it, i.e., $\text{INDSET} \leq_p 2\text{SAT}$
- ☐ NP-hard
- ☒ In the class EXP
- ☒ Solvable in polynomial time

Your answer is correct.

The problem **3SAT** is:

Scegli una o più alternative:

- ☒ In the class **EXP**.
- ☒ such that **INDSET** can be reduced to it, i.e., $\text{INDSET} \leq_p \text{3SAT}$
- ☒ **NP**-hard
- ☐ Computable in polynomial time

The problem of deciding if a 3CNF is satisfiable is

Scegli una o più alternative:

- ☐ Undecidable, but **NP**-hard.
- ☒ In the class **EXP**
- ☐ In the class **P**
- ☒ **NP**-complete

Turing Machines:

Scegli una o più alternative:

- ☐ Always work in a time bounded by a total function f
- ☒ Can sometime work in polynomial time
- ☒ Are such that the presence of multiple tapes changes what can be computed in linear time.
- ☒ Can simulate any other computational model with polynomial overhead

Turing Machines:

Scegli una o più alternative:

- ☐ All work in time bounded by an exponential function
- ☐ All work in time bounded by any function
- ☐ Are such that certain problems that can be computed in time n^2 cannot be computed in time n^3
- ☒ Can contain an arbitrary number of work tapes

Turing Machines:

Scegli una o più alternative:

- ☒ Are such that certain problems that can be computed in time 2^n cannot be computed in time n^2
- ☒ Can work on alphabets of arbitrary but fixed size
- ☐ All work in time bounded by a computable function
- ☐ All work in time bounded by a polynomial function

Turing Machines:

Scegli una o più alternative:

- ☒ Are such certain problems that can be computed in time 2^n cannot be computed in time n , despite it's not known whether P is equal to NP
- ☒ Can all be simulated by one Turing machine
- ☐ Can have infinitely many tapes
- ☐ All work in exponential time

The universal Turing machine:

Scegli una o più alternative:

- ☒ Is an essential ingredient of in the proof of existence of uncomputable problems.
- ☐ Works in polynomial time.
- ☐ Can simulate every Turing machine, but not itself
- ☒ Can simulate every Turing machine, with a polynomial overhead.

Which ones of the following inclusions between complexity classes is coherent with the current knowledge (even if not proved, yet)?

Scegli una o più alternative:

- ☐ $EXP \subseteq NP$
- ☒ $NP \subseteq EXP$
- ☐ $EXP \subseteq P$
- ☒ $P \subseteq EXP$

Let f, g be the functions defined as $f(n) = 10^3 n \log n$ and $g(n) = \frac{n^2}{10^5 \log n}$.

Scegli una o più alternative:

- ☐ $f \in \Omega(g)$
- ☐ $f \in \Theta(g)$
- ☒ $f \in O(g)$

$$\lim_{n \rightarrow +\infty} \frac{10^3 n \log n}{\frac{n^2}{10^5 \log n}} = \lim_{n \rightarrow +\infty} \frac{10^8 n \log^2 n}{n^2} = 0$$

$f \in O(g)$

Let f, g be the functions defined as $f(n) = 10^3 \frac{n}{\log n}$ and $g(n) = 10^{-3} \frac{\log n}{n}$.

Scegli una o più alternative:

- ☒ $f \in \Omega(g)$
☐ $f \in O(g)$
☐ $f \in \Theta(g)$

$$\lim_{n \rightarrow +\infty} \frac{10^3 \frac{n}{\log n}}{10^{-3} \frac{\log n}{n}} = 10^6 \frac{n}{\log n} \cdot \frac{n}{\log n} = \frac{n^2}{\log n} = +\infty$$

$f \in \Omega(g)$

Let f, g be the functions defined as $f(n) = 10^{-7} n^4 \log n$ and $g(n) = \frac{10^2}{\log n} n^5$.

Scegli una o più alternative:

- ☒ $f \in O(g)$
☐ $f \in \Omega(g)$
☐ $f \in \Theta(g)$

$$\lim_{n \rightarrow +\infty} \frac{10^{-7} n^4 \log n}{\frac{10^2}{\log n} n^5} = \lim_{n \rightarrow +\infty} \frac{n^4 \log^2 n}{10^9 n^5} = 0$$

$f \in O(g)$

Let f, g be the functions defined as $f(n) = 2^{10} n^2 \log^2 n$ and $g(n) = 10^{-3} n^3 \log n$.

Scegli una o più alternative:

- ☒ $f \in O(g)$
☐ $f \in \Theta(g)$
☐ $f \in \Omega(g)$

$$\lim_{n \rightarrow +\infty} \frac{2^{10} n^2 \log^2 n}{10^{-3} n^3 \log n} = \lim_{n \rightarrow +\infty} \frac{2^{10} \cdot 10^3 \cdot \log n}{n} = 0$$

$f \in O(g)$

Let f, g be the functions defined as $f(n) = 10^{-8} n^3 \log^{-3} n$ and $g(n) = \frac{10^5}{\log^3 n} n^3$.

Scegli una o più alternative:

- ☒ $f \in \Omega(g)$
- ☐ $f \notin \Theta(g)$
- ☒ $f \in O(g)$

$$\lim_{n \rightarrow +\infty} \frac{10^{-8} n^3 \log^{-3} n}{\frac{10^5}{\log^3 n} n^3} = \lim_{n \rightarrow +\infty} \frac{1}{10^{13}} = \frac{1}{10^{13}} \quad f \in \Theta(g)$$

Suppose a language $\mathcal{L}$ is in **EXP** but not in **P**. Then:

Scegli una o più alternative:

- ☐ $\mathcal{L}$ is necessarily **NP**-complete.
- ☒ The classes **NP** and **P** are different. ✗
- ☒ There could be a nondeterministic polytime TM computing $\mathcal{L}$ ✓
- ☐ $\mathcal{L}$ can be computed in polynomial time.

NOTE: THIS QUESTION IS FROM 2020

(PROF. GABRIELLI)

Your answer is incorrect.

THIS IS A REFERENCE TO NON DETERMINISTIC TM

Le risposte corrette sono: $\mathcal{L}$ can be computed in polynomial time., There could be a nondeterministic polytime TM computing $\mathcal{L}$

Suppose a language $\mathcal{L}$ is in **EXP** but not in **P**. Then:

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Scegli una o più alternative:

- ☒ There could be a nondeterministic polytime TM computing $\mathcal{L}$ ✓
- ☐ $\mathcal{L}$ is necessarily **NP**-complete.
- ☐ $\mathcal{L}$ can be computed in polynomial time.
- ☒ The classes **NP** and **P** are different. ✗

Your answer is incorrect.

ND-TM

Le risposte corrette sono: $\mathcal{L}$ can be computed in polynomial time., There could be a nondeterministic polytime TM computing $\mathcal{L}$

Suppose a language $\mathcal{L}$ is both undecidable and **NP**-hard. Then:

Scegli una o più alternative:

- ☒ There cannot be any nondeterministic Turing machine deciding $\mathcal{L}$
- ☐ The classes **P** and **NP** are necessarily equal.
- ☐ The classes **P** and **NP** are necessarily different.
- ☒ The language $\mathcal{L}$ is strictly harder to decide than SAT



Your answer is correct.

Suppose a language $\mathcal{L}$ is both undecidable and not in **NP**. Then:

Scegli una o più alternative:

- ☐ The classes **P** and **NP** are necessarily equal.
- ☐ The classes **P** and **NP** are necessarily different.
- ☒ The language $\mathcal{L}$ is strictly harder to decide than 3SAT
- ☒ There cannot be any deterministic Turing machine deciding $\mathcal{L}$



Your answer is correct.

The notion of PAC-learnable concept class:

Scegli una o più alternative:

- ☐ Cannot be reached when the underlying concept class is a conjunctions of literals.
- ☒ Is such that the accuracy and confidence errors are taken into account independently.
- ☒ Implicitly refers to the underlying representation class.
- ☐ Is such that the output concept needs to have probability of error ε , in all possible cases



Your answer is correct.

Exercise 2.12. For each of these languages:

- $L_1 := \{\ulcorner M \urcorner \mid M \text{ rejects every input}\},$
- $L_2 := \{\ulcorner M \urcorner \mid M \text{ accepts every input in exactly 42 steps}\},$
- $L_3 := \{\ulcorner M \urcorner \mid M \text{ accepts more than 42 different inputs}\},$
- $L_4 := \{\ulcorner M \urcorner \mid M \text{ rejects the input } 0^{42}\},$
- $L_5 := \{\ulcorner \ulcorner M \urcorner, x \urcorner \mid \text{on input } x, M \text{ writes on the 42th cell of the tape}\},$
- $L_6 := \{\ulcorner M \urcorner \mid M \text{ has more than 42 states}\},$
- $L_7 := \{\ulcorner M \urcorner \mid M \text{ accepts a finite language}\},$
- $L_8 := \{w \in \{0,1\}^* \mid \text{if } w = \ulcorner M \urcorner \text{ then } M \text{ accepts 101 or rejects 101}\}.$

say if it is (1) semantic, (2) trivial, (3) decidable.

Exercise 2.13. Are the following languages decidable?

- $L_1 := \{\ulcorner M \urcorner \mid M \text{ terminates with output 100 on input 001}\},$
- $L_2 := \{\ulcorner M \urcorner \mid M \text{ is a Turing machine with at most five states}\},$
- $L_3 := \{\ulcorner \ulcorner M \urcorner, x \urcorner \mid \text{on input } x, M \text{ does not terminate before three computation steps}\},$
- $L_4 := \{\ulcorner \ulcorner M \urcorner, x \urcorner \mid \text{on input } x, M \text{ terminates before three computation steps}\},$
- $L_5 := \{\ulcorner \ulcorner M \urcorner, x \urcorner \mid \text{on input } x, M \text{ terminates after three computation steps or more}\}$
- $L_6 := \{w \in \{0,1\}^* \mid \text{if } w = \ulcorner M \urcorner \text{ then } \mathcal{L}(M) \text{ contains } 0011\}.$

S : SEMANTIC T : TRIVIAL D : DECIDABLE
 NS : NON-SEMANTIC NT : NON TRIVIAL UD : UNDECIDABLE

L_1 $S, NT \rightarrow UD$
 L_2 $NS, NT \rightarrow D$
 L_3 $S, NT \rightarrow UD$
 L_4 $S, NT \rightarrow UD$
 L_5 $NS, NT \rightarrow D$
 L_6 $NS, NT \rightarrow D$
 L_7 $S, NT \rightarrow UD$
 L_8 $S, T \rightarrow D$

L_1 $S, NT \rightarrow UD$
 L_2 $NS, NT \rightarrow D$
 L_3 $NS, NT \rightarrow D$
 L_4 $NS, NT \rightarrow D$
 L_5 $NS, NT \rightarrow UD$
 L_6 $S, NT \rightarrow UD$

A problem $\square$ is said to be **EXP**-complete if it is in **EXP** and any problem in **EXP** can be reduced, in polynomial time, to $\square$. Suppose a friend of yours comes to you claiming that he found an **EXP**-complete problem which is also in **NP**. What can you say?

Scegli una o più alternative:

- ☒ That your friend is perhaps right, but that, if right, he would have proved $P \neq NP$. ✓
- ☐ That your friend is certainly wrong, because such a problem cannot exist.
- ☐ That your friend is certainly right, because finding such a problem is very easy.
- ☐ That your friend is perhaps right, but that, if right, he would have proved $P = NP$.

The problem of deciding whether an undirected graph G contains a clique of size equal to the size of G is:

Scegli una o più alternative:

- ☐ Undecidable, and **NP**-hard.
- ☒ In the class **EXP** ✓
- ☐ Decidable in polynomial time
- ☐ Not in the class **NP**

Which of the following sets can be easily encoded as binary strings?

Scegli una o più alternative:

- ☒ Strings on an alphabet of size 10^2 ✓
- ☒ Strings on an alphabet of size 10^4 ✓
- ☐ Vectors of complex numbers
- ☒ Vectors of rational numbers ✓

15/06/2026

The following sets can be encoded as binary strings:

Scegli una o più alternative:

- ☐ The set of triples of real numbers, i.e., $\mathbb{R} \setminus \text{time} \mathbb{R} \times \mathbb{R}$.
- ☐ The set of complex numbers, i.e. $\mathbb{C}$.
- ☒ The set of pairs of rational numbers, i.e., $\mathbb{Q} \times \mathbb{Q}$.

The problem of checking whether two Turing machines are such that the former has more states than the latter is:

Scegli una o più alternative:

- ☐ Not in **NP**
- ☒ In the class **EXP**
- ☒ Decidable in polynomial time
- ☐ Undecidable.

Which ones of the following inclusions between complexity classes is compatible with our current knowledge (i.e. maybe we do not know if this is true, but it could be)? Here, **NPC** stands for the class of all **NP**-complete problems.

Scegli una o più alternative:

- ☐ **P = EXP**.
- ☒ **P $\supseteq$ NPC**.
- ☒ **NP $\subseteq$ NPC**.
- ☐ **PSPACE $\subseteq$ L**.

A friend of yours claims he designed an algorithm solving the **SAT** and working in polynomial space. You can safely conclude that:

Scegli una o più alternative:

- ☐ If your friend is right, then **P = NP**.
- ☒ Your friend's algorithm might be correct, but one has of course to check the details.
- ☒ Even if your friend is correct, **SAT** would stay **NP**-complete.
- ☐ Your friend must be right, because all algorithms solving **SAT** works in polynomial space.

The representation class of closed intervals in the real line $\mathbb{R}$:

Scegli una o più alternative:

- ☒ Is efficiently PAC-learnable.
- ☐ Has a VC-dimension equal to that of axis-aligned rectangles in $\mathbb{R}^2$.
- ☒ Has VC-dimension equal to 2
- ☐ Has infinite VC-dimension.